Index

Page numbers in **bold** refer to main entries.

3-D modeling software 177
3-D printing **86–87**

A

acceleration
  electric and hybrid cars 46, 47
  trains 52
accelerometers 207
acrylic 126
actuators 18, 172
additives, food 226, 227
aeration lanes 13
aerobic decomposition 121
aeroponics 218
aggregate 76
agriculture *see* farming
agrobacterium 228
AI *see* artificial intelligence
ailerons 62
air-conditioning **117**
air cycle machines 63
air pollution 46
air pressure 54, 63, 118, 136, 138
air traffic control 48–49
airbags 69
aircraft
  airplanes 38, 39, **62–63**
  drones 66–67, 220, 221
  engines 60–61
  helicopters 64–65
  military 49
airfoils 54, 62
airplanes 38, 39, **62–63**
  food 227
alarm systems **124–125**
algorithms 132, 133, 168, 169
alloys 72, 73
alternating current (AC) 16, 17, 18, 52, 139, 182
alternators 16, 52
ALU (arithmetic logic unit) 165
amplification 139, 185
amplitude 137, 183
amplitude modulation (AM) 181, 182
anaerobic digestion 214
analogue radio 183
analogue television 189
analogue-to-digital chips 161
analogue-to-digital converters (ADC) 140, 153, 155, 158
Anderson, Robert 47
angle of attack 62
animals
  farming **214–215**, 220, 229
  fibers 126
anodes 32, 33, 34, 111, 190, 234
anodizing 75
antennas
  cell phones 196
  radar 48, 49
  radio signals 180, 181, 182, 195, 202
  satellites 192, 193
  spacecraft 68
  television 189
antithrombin 229
Apollo missions 35
applications, computer 168
apps 132, 133, 191, 206, 207
aquaponics **219**
arch bridges 94
arches **88–89**
architecture
  arches and domes 88–89
  bridges 94–95
  skyscrapers 94–95
armatures 18, 19
arms, prosthetic 240
ARPANET 200
artificial intelligence (AI) 47, **170–171**
artificial neuron network 170
astronomy 48
atomic bonds 33
atoms, splitting 24
atria 232, 233
attraction, in electric motors 18, 19
audio signals 138–139, 140–141, 158, 187
auditory nerve 243
augmented reality **177**
automatic transmission, in cars 44
autonomous robots 175
autosomes 244
axles 38
azimuth thrusters 57

B

bacteria 15, 137, 219
  food preservation 223, 224, 225
  genetic modification 228, 229
  refrigeration 116
  water supply 12, 13
ballast tanks 58
bandwidth 203
banknotes 148
bases, sequence of 245
basic oxygen converters (BOC) 72, 73
batteries 32–33
  electric cars 46–47
beam bridges 94
Bell, Alexander Graham 184, 185
bending, of metals 75
Berners-Lee, Sir Tim 199
bicycles **40–41**
  carbon fiber 81
  dynamos 17
  wheels 39
bimetallic thermostats 109, 112
binary numbers 158, 159
  computers 166, 168, 169
  digital electronics 160, 161
  digital images 153, 154
  digital sound 140, 141, 183
binoculars 142, **143**
bioenergy **31**
biofuel 31
biogas 215
biomass, as an energy source 31
biomethane 215
biometric locks 123, 133
bionic eyes 243
biopreservation 225
bits (binary digits) 158, 159
bits (drills) 97
biventricular pacemakers 232
blades
  drones 66–67
  helicopter 64–65
  turbine 26
blast furnaces 72
bleed air 63
blowout preventers 91
blue light 208
Bluetooth 167, 205, 206
boats **54–57**
boilers 108, 109
bone 80
boreholes 91
boring machines, tunnel 96–97
boron 160
botnet 200, 201
botulism 225
bow thruster 57
brain
  implants **242–243**
  nerve signals 240
  and virtual reality 176, 177
brakes 41, 45
  elevators 100
  regenerative 47
brass 73
breathable fabrics 127
bridges **94–95**
broadcasting
  radio 180, 181, **182–183**
  television **188–189**, 190–191
bronze 73
browsers, web 198, 199
Brunelleschi, Filippo 88
bubble cap trays 14
buckyballs 84
building materials *see* materials
buoyancy 55, 58
buses 164
bytes 158, 159

C

cabin zones, airliner 63
cable-stayed bridges 94
cables
  electric 185
  fiber-optic 185, 187, 196, 197
  television 188–189